

- There are 4 Questions for 15 marks.
- Write your answers neatly, legibly and logically. Keep your proofs thorough. Your proofs must follow from the basics discussed in class.
- You do not need any calculators.
- Good luck!

1. [4] Suppose that over a certain region of space the electrical potential is given by $V(x, y, z) = 5x^2 - 3xy + xyz$.
 - (a) What is the rate of change of the potential at point $(1, 0, 2)$ along the direction of $(\mathbf{i} + \mathbf{j} - \mathbf{k})/\sqrt{3}$?
 - (b) In which direction does V change the most rapidly at $(1, 0, 2)$?
2. The length l , width w , and height h of a box are changing with time. At $t = 0$, the dimensions are $(l, w, h) = (1m, 1m, 1m)$ and are increasing at the rate $1m/s$, $2m/s$ and $4m/s$ respectively for l, w, h .
 - (a) [2] What is the rate at which the volume is changing at $t = 0$?
 - (b) [2] Find the approximate volume at $t = 0.01s$.
3. [3] Consider a space curve in $\mathbb{R}^3$ of the form

$$\mathbf{r}(t) = (t^3 + t^2 \quad t^2 + t \quad t)$$

for $t \in \mathbb{R}$. Does this curve lie on a plane?

4. [4] A rectangle with width L and length W is cut into four smaller rectangles by two lines parallel to the sides. As a function of the position of these lines, let S represent the sum of squares of the areas of smaller rectangles. What are the maximum and minimum values of S ?

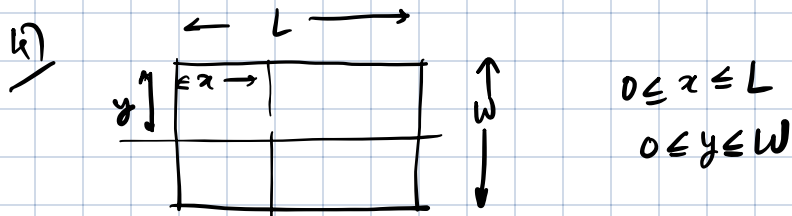
1) $V(x, y, z) = 5x^2 - 3xy + xyz$

a) Gradient is $\nabla V = \begin{bmatrix} 10x - 3y + yz \\ -3x + xz \\ xy \end{bmatrix}$ at $x=1, y=0, z=2$ $\nabla V = \begin{bmatrix} 10 \\ -1 \\ 0 \end{bmatrix}$

Rate of change is the inner product of gradient with the direction vector : $9/\sqrt{3} = 3\sqrt{3}$.

b) Direction of most rapid change is +ve gradient (for increase) & negative gradient (for decrease)

$$\pm \begin{bmatrix} 10 \\ -1 \\ 0 \end{bmatrix} / \sqrt{101}$$



then $S(x, y) = x^2y^2 + (L-x)^2y^2 + x^2(L-y)^2 + (L-x)^2(L-y)^2$

$$\begin{aligned} \frac{\partial S}{\partial x} &= 2xy^2 - 2(L-x)y^2 + 2x(L-y)^2 - 2(L-x)(L-y)^2 \\ &= 4xy^2 + 4x(L-y)^2 - 2Ly^2 - 2L(L-y)^2 \\ &= 4[y^2 + (L-y)^2] \left[x - \frac{L}{2}\right] \end{aligned}$$

By symmetry, $\frac{\partial S}{\partial y} = 4[x^2 + (L-x)^2] \left[y - \frac{L}{2}\right]$

the gradient vanishes only when $x = \frac{L}{2}$, $y = \frac{L}{2}$ leaving only one possible local min/max.

Now, $\frac{\partial^2 S}{\partial x \partial y} = 4 \left[x - \frac{L}{2}\right] [2y - 2(L-y)] = 16 \left[x - \frac{L}{2}\right] \left[y - \frac{L}{2}\right]$

$$\frac{\partial^2 S}{\partial x^2} = 4[y^2 + (L-y)^2]$$

Hessian at $x = \frac{L}{2}$, $y = \frac{L}{2}$ is

$$\begin{bmatrix} 2L^2 & 0 \\ 0 & 2L^2 \end{bmatrix}$$

Which has +ve eigenvalues. Therefore $x = \frac{L}{2}, y = \frac{L}{2}$ is a local minima. The value of S at $x = \frac{L}{2}, y = \frac{L}{2}$ is

$$S = L^4/4.$$

Now, the value at the boundary is

$$\text{either } L^2(y^2 + (L-y)^2) \quad (\text{when } x=0 \text{ or } x=L)$$

$$\text{or } L^2(x^2 + (L-x)^2) \quad (\text{when } y=0 \text{ or } y=L)$$

In either case, the minimum value on boundary is $L^4/4$ and maximum value on boundary is L^4 (min: when $x=L/2$ & $y=0$ or $y=L$
max: when at corners)

Thus, overall min value is $L^4/4$ when $x=L/2, y=L/2$

Overall max is L^4 when $x=0, y=0$.

2) $V = lwh$

$$(a) \frac{dV}{dt} = \frac{\partial V}{\partial l} \frac{dl}{dt} + \frac{\partial V}{\partial w} \frac{dw}{dt} + \frac{\partial V}{\partial h} \frac{dh}{dt}$$

$$= wh(1) + lh(2) + lw(4) = 7 \text{ m}^3/\text{s} \quad \text{at } l=w=h=1\text{m}$$

$$(b) V(0.01) \approx V(0) + (0.01)V'(t) = 1 + 0.07 = 1.07.$$

3) Note that $(0,0,0)$ lies on the curve. Suppose the curve is contained in a plane $ax+by+cz=0$.

then we must have $a(t^3+t^2) + b(t^2+t) + ct = 0$ for all t .

$$\text{i.e. } a=0, a+b=0, b+c=0 \Rightarrow a=b=c=0.$$

So the curve does not lie on a plane.